A Project Report

On

ENERGY-AWARE AI INFERENCE FOR DATA CENTRE ENERGY MANAGEMENT: SMALL VS LARGE LANGUAGE MODELS

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Certificate

This is to certify that the project report entitled "ENERGY-AWARE AI INFERENCE FOR DATA CENTRE ENERGY MANAGEMENT: SMALL VS LARGE LANGUAGE MODELS (NLI TASK)" submitted by Rudresh Asagodu (ID No. 2023B5A80772H) in partial fulfillment of the requirements of the course INSTR F376: Design Project, embodies the work done by him under my supervision and guidance.

Date: October 13, 2026

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1 INTRODUCTION

Data centres consume vast amounts of electricity, with an increasingly large share dedicated to running artificial intelligence workloads. Every inference request draws GPU power for the duration of execution. However, not every task requires the largest general-purpose model available. Structured or specialized tasks—such as natural language inference (determining the logical relationship between sentence pairs)—can often be handled efficiently by task-specific models that are a fraction of the size of large language models (LLMs). Directing requests to models appropriately sized for the task enables data centres to achieve the same useful computational output while substantially reducing energy consumption.

This project fits into a broader initiative on data centre energy management with colocated renewable energy, aligning computational demand with on-site solar generation, storage, and grid signals. While the supply-side focuses on power delivery, this report addresses the demand-side: minimizing the power required by AI inference workloads

Building upon foundational work comparing small and large language models, this study evaluates Natural Language Inference (SNLI) on real hardware, directly measuring GPU energy and performance metrics.

This report describes:

1. the reference paper and how it fits into the larger project (Section 2);
2. the task, the three models, the hardware and the way energy was measured (Section 3);
3. the quality results of the three models (Section 4);

6. the conclusions and the plan for the rest of the project (Section 7).

2 Reference Paper and Project Context

2.1 The paper in brief

Prior literature investigates whether task-specific small language models (SLMs) can match general-purpose large language models (LLMs) at a fraction of energy and emissions. While small models perform competitively on structured classification tasks, complex reasoning and generative tasks often present distinct trade-offs. Furthermore, many published emission figures rely on theoretical estimates scaled from aggregate data rather than direct hardware measurements.

2.2 What this project adds

Three things were done differently from the paper:

irectly Measured Energy: GPU power is sampled directly during inference and integrated over time to compute exact energy consumption in joules.

Rigorous Replication: Evaluations are run across multiple trials (e.g., 5 repeated runs) to ensure stability and statistical significance.

Task-Specific Evaluation: Focuses on Natural Language Inference (SNLI) to test logical reasoning capabilities between compact and large models.

2.3 Project plan

The work was planned over five months. This report covers progress within Month 1.

Month	Goal
1	Reproduce a paper task and measure real SLM/LLM inference energy
2	Build a confidence-based SLM→LLM router
3	Model the data-centre grid connection and a co-located battery; add price and carbon signals
4	Jointly optimize model routing and battery dispatch (heuristic, optimization, learning-based)
5	Data-centre-scale simulation, evaluation and write-up

Table 1: Five-month project plan.

3 Experimental Setup

3.1 Task and dataset

The benchmark task is Natural Language Inference (SNLI), which requires reading two sentences (a premise and a hypothesis) and determining their logical relationship: Entailment (must be true), Contradiction (cannot be true), or Neutral (maybe true). Unclear pairs were filtered out, and a balanced subset of 1,000 sentence pairs was randomly sampled for evaluation.

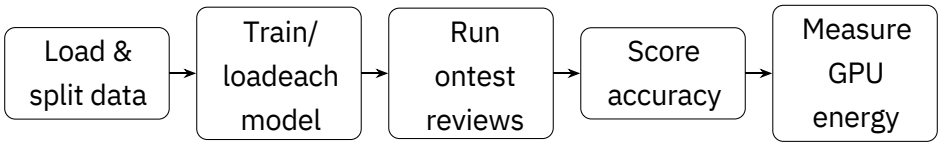


Figure 1: Experimental workflow.

3.2 Models

- DeBERTa-v3-base (Small Model): A compact model featuring 184 million parameters, fine-tuned specifically for the NLI task, processing both sentences together and outputting the classification directly. Memory utilization stood at 0.69 GB.
- Mistral-7B-Instruct (Large Model): A general-purpose LLM with 7 billion parameters, prompted zero-shot with instructions and 6 examples, outputting 'Yes', 'No', or 'Maybe'. To fit within the GPU memory limits, it was quantized to 4-bit (NF4), utilizing 3.74 GB of memory.

3.3 Hardware and energy measurement

All experiments were executed on a Google Colab GPU environment. GPU energy counters were queried before and after execution to capture exact energy consumption without startup overhead (discarding warm-up queries). Each complete evaluation was repeated 5 times to ensure consistency.

3.4 Metrics

- Accuracy: Percentage of correct predictions against ground-truth labels.
- Latency (Time per query): Duration required to process a single query.
- Energy per query: Total integrated GPU energy divided by the 1,000 test pairs.

Carbon Emissions CO2: Calculated based on energy consumption assuming an emission factor of 0.7 kg CO2 per kWh

4 Results: Model Quality

On the 1,000-sample SNLI test subset evaluated across 5 repeated runs, DeBERTa-v3-base achieved an accuracy of 92.4%, whereas Mistral-7B (4-bit) achieved 76.8%.

5 Results: Energy and Performance

The performance comparison demonstrates a decisive advantage for the task-specific small model across all measured dimensions.

Metric	DeBERTa (Small)	Mistral-7B (Large)
Accuracy	92.4% PDF	76.8% PDF
Time per query	25 ms PDF	572 ms PDF
Energy per query	1.37 J PDF	39.3 J PDF
CO2 per query*	0.00027 g PDF	0.00764 g PDF

Key comparative findings include:

Energy Consumption: Mistral-7B consumed 39.3 J per query compared to just 1.37 J for DeBERTa, meaning the large model required approximately 29 times more energy per query.

Latency: DeBERTa completed inferences in an average of 25 ms, while Mistral required 572 ms, making Mistral approximately 23 times slower.

Carbon Footprint: Carbon emissions scaled proportionally with energy, with Mistral producing 0.00764 g of CO2 per query versus 0.00027 g for DeBERTa.

6 Conclusion and Future Works

This study evaluated the trade-offs between a task-specific small language model (DeBERTa-v3-base) and a large language model (Mistral-7B) on the Natural Language Inference (SNLI) task. The small model outperformed the large model on all fronts—achieving higher accuracy (92.4% vs 76.8%), operating faster (25 ms vs 572 ms), and consuming vastly less energy (1.37 J vs 39.3 J). These findings validate that deploying appropriately sized models is a powerful mechanism for data centre energy conservation.

Future Work:
Develop a confidence-based adaptive routing framework to delegate straightforward queries to small models and escalate only low-confidence cases to LLMs.
Integrate carbon intensity signals and model energy profiles into a colocated battery and grid-dispatch optimization model for data centres.
Expand evaluations to additional task categories (e.g., code generation or complex reasoning) and scale simulations to data-centre levels.

7 REFERENCES

1. Garg, Gaba, Muthirayan and Chowdhury, “Emissions and Performance Trade-off Between Small and Large Language Models,” Plaksha University, FLLM 2025.